Project Proposal - Rat Prefrontal Cortex Layer V Pyramidal Neuron

The paper involves studying the layer V pyramidal neurons in a rat PFC, and modifying the M-current using the three drugs XE991, retigabine, and QO58L. They have recorded the effect on the M-current and spike frequency as figures in the paper, comparing the effects of the three drugs against a control.

One model that can be used would be based on the minimal Hodgkin-Huxley models of regular-spiking cortical pyramidal neurons, which may give us a $4D$ or $5D$ dynamical system.

Then, our extension could be investigating how much M-current suppression is required to change the neuronal dynamics (e.g., for the XE991 drug). For example, if we had the variable I_M which is used in the regular-spiking model, we may model it to depend on the blocking factor $\rho \in [0, 1]$, then conduct bifurcation analysis on the applied current and ρ, i.e., on (I_{app}, ρ). Our physiological question would then be something like:

> **ⓘ Physiological Context**
>
> How does pharmacological suppression of the Kv7/M-current shift the threshold separating quiescent and repetitively firing behaviour in layer V PFC pyramidal neurons?

Another extension would be to make retigabine more interesting mathematically, as it modifies the voltage dependence of the M-channel activation, including the shifting activation in the hyperpolarising direction.

Then, we can represent the drug by a shift in the M-channel activation, and investigate whether that moves the bifurcation boundary symmetrically or in qualitatively different ways. We can either do this independently of our change in the XE991 drug model assumption, or conduct both at the same time afterwards as well to see what bifurcations can happen.

One caveat is that the cells the study is based on is not exactly modelled by the minimal model for the more generic regular-spiking cortical pyramidal model mentioned earlier. In this case, we may want to adapt or calibrate a minimal cortical Hodgkin-Huxley model to this data, and tune a small number of parameters so that our baseline model approximately matches the experimental data, while being sure not to overfit too many parameters.

This will be the proposal for now, I can add some more of the technical details after further reading and after this has been greenlit.